Md Shamiul Islam Riyad

Applied AI · RAG and LLM Applications · CSE Undergraduate

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SUMMARY

CSE undergraduate who builds working AI applications: RAG pipelines with cited answers, LLM assistants with safety checks and fallbacks, and Bengali vision and voice tools on the Gemini and Claude APIs. Writes step-by-step documentation and offline tests for each build; 7 hackathons in 2026. Seeking the BIAI Applied AI Internship to build demos and learning content, test AI tools and learn from senior builders.

EDUCATION

BSc in Computer Science and Engineering, United International University (UIU), Dhaka 2023 – 2027 (expected)

AI coursework: k-nearest neighbours, k-means, A* search and a CSP solver, implemented from scratch in Python.

APPLIED AI PROJECTS

ragkit-dev – open-source RAG toolkit for learning and reuse github.com/shamiulriyad/rag-learning

Python, LangChain, Qdrant, Hugging Face embeddings, Gemini, FastAPI, Streamlit. Personal project.

- Split the RAG pipeline into 11 single-purpose modules (extract, clean, chunk, embed, retrieve, generate…) so each step can be read, tested and swapped; answers cite the source chunk and PDF page.
- Packaged it with a CLI, FastAPI service, Streamlit test UI and an npm generator that guides new users through setup.
- Wrote 290+ tests and an offline GitHub Actions pipeline that needs no API keys; documented setup and every step.

FactoryBrain – AI operations platform for garment factories (English and Bangla) github.com/shamiulriyad/FactoryBrain

Python, FastAPI, Qdrant, ASP.NET Core 9, Next.js; OpenAI, Anthropic or Gemini; 174 xUnit and 57 Playwright tests. Personal project.

- Built cited Q&A over factory manuals through a Python RAG service that answers from retrieved chunks only.
- Added AI agents whose medium- and high-risk runs wait in a human approval queue, and garment defect detection with a vision-language model plus inspector review.

QueueStorm Investigator – LLM support copilot for digital finance github.com/shamiulriyad/sust_preli

ASP.NET Core 8, Claude API, Docker. bKash SUST CSE Carnival 2026 hackathon.

- Checks a complaint in English, Bangla or Banglish against transaction history and returns one JSON decision: verdict, matching transaction, routing team and draft reply.
- Falls back to a rule-based engine if Claude fails or times out; every reply passes schema validation and a safety filter that blocks PIN/OTP requests and refund promises.

Chokh – AI narration for visually impaired users, in Bengali Live demo

React, ASP.NET Core, Gemini vision, Web Speech API, Docker. ML Bangladesh Hackathon 2026.

- Deployed web app: sends a camera frame to Gemini every ~2.5 s and speaks hazards first, then position and text.
- Wrote the Bengali system prompt to admit uncertainty rather than invent objects, and to refuse medical advice.

NogorSathi AI – Gemma 4 commuter-agent prototype github.com/shamiulriyad/NogorSathi_UIU_GemmaHackathon

JavaScript, HTML, CSS. Team project, Build with Gemma @ Bangladesh 2026.

- Built the demo interface that shows each tool call (geocoding, routes, flood and safety checks) and explains why one Dhaka route beats the others, in Bangla and English.

OTHER PROJECTS

- **rag-kit:** multi-tenant RAG platform; ASP.NET Core gateway, FastAPI RAG service, one Qdrant collection per knowledge base.
- **AI Study Assistant:** ask by text or voice, get a simple Bangla answer from Gemini, read aloud in the browser.
- **ShilpoHubBD:** team of 5; wrote 790 of 964 backend commits (ASP.NET Core, PostgreSQL), merged 96 pull requests.

TECHNICAL SKILLS

AI and LLMs: Gemini and Claude APIs, prompt design, RAG (chunking, embeddings, retrieval, citations), LangChain, Qdrant
Languages: Python, C#, TypeScript, JavaScript, C++, SQL
Frameworks and tools: FastAPI, Streamlit, ASP.NET Core, React, Next.js, PostgreSQL, Docker
Testing and CI: pytest, xUnit, Playwright, GitHub Actions, Git
Currently learning: PyTorch and computer vision fundamentals

ACHIEVEMENTS

- Featured in Google Developer Experts AI Community Highlights, Build with Gemma @ Bangladesh (Jul 2026).
- Top 70 of 700+ teams, Infinity AI BuildFest (Jun 2026); 2nd runner-up, UIU CSE Project Show (Mar 2026).
- 575+ problems solved in 55+ contests (LeetCode 1562, CodeChef 1448); 4th place, UIU Team Contest (Jul 2026).